From Classical harmonic oscillator to quantum:

Mass m attached to a spring of force constant k.

$$F = m\frac{d^2x}{dt^2} = -kx$$

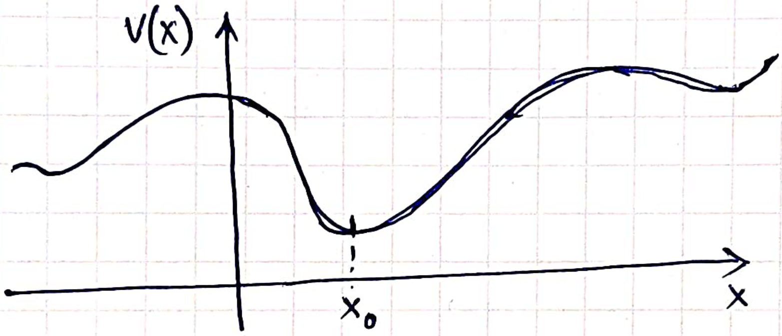
$$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \Rightarrow \frac{d^2x}{dt^2} + \omega^2 x = 0, \quad \omega = \sqrt{\frac{k}{m}}$$

ω → angular frequency

General soln; $x(t) = A\sin\omega t + B\cos\omega t$

potential energy $V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2$

Hamiltonian $\boxed{H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2}$



Arbitrary potential landscape with local minima.

Expand $V(x)$ in a Taylor series in the neighbourhood of a local minimum at $x = x_0$.

$$V(x) = V(x_0) + \left.\frac{dV}{dx}\right|_{x=x_0}(x-x_0) + \frac{1}{2}\left.\frac{d^2V}{dx^2}\right|_{x=x_0}(x-x_0)^2 + \ldots$$

Now $\left.\frac{dV}{dx}\right|_{x=x_0} = 0 \Rightarrow$ Dropping higher order terms, $(x-x_0)$ being small,

$$V(x) \simeq \frac{1}{2}\left.\frac{d^2V}{dx^2}\right|_{x=x_0}(x-x_0)^2$$

Define an effective force constant $k = \left.\frac{d^2V}{dx^2}\right|_{x=x_0(=0)} \Rightarrow V(x) \simeq \frac{1}{2}kx^2$

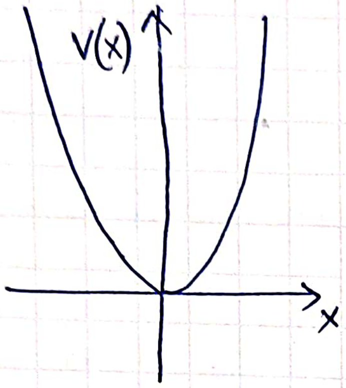
Any potential resembles a harmonic oscillator potential in the neighbourhood of a local minimum

The time independent Schrodinger eqn for a quantum harmonic oscillator

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2x^2\psi = E\psi}$$

$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$

where $[\hat{x}, \hat{p}] = i\hbar$



$$V(x) = V(-x)$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Let $x' = -x$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(-x')}{dx'^2} + \underbrace{V(-x')}_{V(x')}\psi(-x') = E\psi(-x')$$

Now drop the prime !

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(-x)}{dx^2} + V(x)\psi(-x) = E\psi(-x)$$

No degeneracy in 1D $\Rightarrow$ $\psi(-x)$ is linearly dependent on $\psi(x)$

$\Rightarrow \psi(-x) = c\,\psi(x) \Rightarrow \boxed{\psi(-x) = \pm\psi(x)}$

[replace x with $-x$

$\Rightarrow \psi(x) = c\,\psi(-x) = c^2\psi(x)$

$\Rightarrow c = \pm 1$]

Every eigenfunction is either an even or odd function
ground state has no nodes $\Rightarrow$ even function

This ofcourse does not imply that every possible quantum state for a harmonic oscillator is either even or odd. One can always construct a linear superposition of even and odd eigenstates which is neither odd nor even!

$$-\frac{\hbar^2}{2m}\frac{d^2\varphi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\varphi(x) = E\varphi(x)$$ 3

Introduce a dimensionless variable, $x = a\xi$, where ξ is dimensionless and $[a] = L$

Express a in terms of $m, \omega, \hbar$

constant with dimension of length.

Note that $\frac{\hbar^2}{ma^2}$ and $m\omega^2a^2$ are dimensionally the same.

$$\Rightarrow \frac{\hbar^2}{ma^2} \equiv m\omega^2a^2 \Rightarrow a^2 = \frac{\hbar}{m\omega} \Rightarrow x = \sqrt{\frac{\hbar}{m\omega}}\,\xi$$

$$\Rightarrow -\frac{\hbar^2}{2ma^2}\frac{d^2\varphi}{d\xi^2} + \frac{1}{2}m\omega^2a^2\xi^2\varphi(x) = E\varphi(x)$$

Note: $\frac{\hbar^2}{ma^2} = \frac{\hbar^2}{m}\cdot\frac{m\omega}{\hbar} = \hbar\omega$, and $m\omega^2a^2 = m\omega^2\cdot\frac{\hbar}{m\omega} = \hbar\omega$

$$\Rightarrow -\frac{d^2\varphi}{d\xi^2} + \xi^2\varphi(x) = \frac{2E}{\hbar\omega}\varphi(x)$$

We define another dimensionless quantity $\varepsilon = \frac{2E}{\hbar\omega} = \frac{E}{\hbar\omega/2}$

$$\Rightarrow \boxed{E = \frac{\hbar\omega}{2}\varepsilon}$$

$$\Rightarrow \boxed{\frac{d^2\varphi}{d\xi^2} = (\xi^2-\varepsilon)\varphi} \quad \text{---- (1)}$$

For large ξ ($\xi \to \pm\infty$), try $\varphi(\xi) = Ae^{-\xi^2/2}$

$$\frac{d\varphi}{d\xi} = A(-\xi)e^{-\xi^2/2}\ ; \quad \frac{d^2\varphi}{d\xi^2} = A\left[\xi^2e^{-\xi^2/2} - e^{-\xi^2/2}\right]$$

$$\simeq A\xi^2e^{-\xi^2/2} \simeq \xi^2\varphi \quad \text{---- (2)}$$

$\Rightarrow$ If φ is a polynomial of degree n, $\frac{d^2\varphi}{d\xi^2}$ will be of degree $n-2$ while $\xi^2\varphi$ will be of degree $n+2$.

We try a soln of the form $\varphi(\xi) = A\xi^k e^{\alpha\xi^2/2}$

As $|\xi| \to \infty$ $\frac{d^2\varphi}{d\xi^2} \approx \alpha^2\xi^2\varphi$ (keeping the leading term only) ---(3)

comparing (2) and (3) we get $\alpha = \pm 1$. The soln for $\alpha = +1$ is unacceptable as it diverges as $\xi \to \infty$

We finally write the trial soln $\varphi(\xi) = h(\xi)e^{-\xi^2/2}$, where we expect $h(\xi)$ to be a polynomial.

Recall, $\frac{d^2\varphi}{d\xi^2} = (\xi^2 - \varepsilon)\varphi$ (4)

$$\Rightarrow \boxed{\frac{d^2h}{d\xi^2} - 2\xi\frac{dh}{d\xi} + (\varepsilon - 1)h = 0} \quad --- (4)$$

Recall: Harmonic oscillator potential is even; bound states solns must either be even or odd. Since $e^{-\xi^2/2}$ is even, soln $\varphi(\xi)$ will be either even or odd if $h(\xi)$ is even or odd.

Use the series expansion $h(\xi) = \sum_{k=0}^{\infty} a_k \xi^k$ --- (5)

Plug in the series expansion in eqn (4) and collect the coefficients of ξ^j and equate them to zero.

$$\frac{d^2h}{d\xi^2} = (j+2)(j+1)a_{j+2}\xi^j$$

$$-2\xi\frac{dh}{d\xi} = -2ja_j\xi^j$$

$$(\varepsilon-1)h = (\varepsilon-1)a_j\xi^j$$

$$\sum_{j=0}^{\infty}\left[(j+2)(j+1)a_{j+2} - 2ja_j + (\varepsilon-1)a_j\right]\xi^j = 0$$

$$\Rightarrow (j+2)(j+1)a_{j+2} = (2j+1-\varepsilon)a_j$$

$$\Rightarrow \boxed{a_{j+2} = \frac{2j+1-\varepsilon}{(j+2)(j+1)}a_j} \quad j = 0, 1, 2, \ldots$$

start from $j=0$, ie a_0 and calculate $a_2, a_4, a_6, \ldots$ etc.
The resulting soln $a_0 + a_2\xi^2 + a_4\xi^4 + \ldots$ will be an even soln.
Similarly start with $j=1$, ie a_1 and calculate $a_3, a_5, a_7, \ldots$ etc.
The resulting soln $a_1\xi + a_3\xi^3 + a_5\xi^5 + \ldots$ will be an odd soln.

For large j, the recursion relation reduces to,

$$\frac{a_{j+2}}{a_j} \approx \frac{2}{j}$$

On the other side, take the function e^{ξ^2}.

$$e^{\xi^2} = \sum_{j=0}^{\infty} \frac{1}{j!}\left(\xi^2\right)^j = \sum_{j' \text{ even}} \frac{1}{(j'/2)!}\,\xi^{j'} \qquad (j'=2j)$$

The series has coefficients $c_j = \frac{1}{(j/2)!}$ for j even

$$\Rightarrow \quad \frac{c_{j+2}}{c_j} = \frac{(j/2)!}{(j+2)/2\,!} = \frac{2}{j+2} \approx \frac{2}{j} \quad (\text{for large } j)$$

This is the same recursion relation as obtained for $h(u)$.

Therefore $\varphi(\xi) = h(\xi)\,e^{-\xi^2/2} \sim e^{\xi^2} e^{-\xi^2/2} \sim e^{\xi^2/2}$.

The soln diverges. Therefore $h(\xi)$ must be a polynomial and the recursion relation must terminate to get an energy eigenstate.

If $h(\xi)$ is to be a polynomial of degree $j=n$, it must have a nonvanishing a_n and $a_{n+2}=0$

$$\Rightarrow \quad 2n+1-\varepsilon = 0$$

. The soln is even if n is even and the soln is odd if n is odd.

$$\boxed{\varepsilon = 2n+1}, \quad n = 0, 1, 2, \ldots$$

and the polynomial soln $\boxed{h_n(\xi) = a_n\xi^n + a_{n-2}\xi^{n-2} + \ldots}$ $n=0,1,2,\ldots$

The full soln $\varphi_n(\xi) = h_n(\xi)\,e^{-\xi^2/2}$, with energy eigenvalues given by $E = \frac{\hbar\omega}{2}\varepsilon = \frac{\hbar\omega}{2}(2n+1) \Rightarrow \boxed{E_n = \left(n+\tfrac{1}{2}\right)\hbar\omega}$ $n=0,1,2,\ldots$

The energy is quantized and the energy levels are evenly spaced. The ground state energy is $\frac{1}{2}\hbar\omega$.

$h_n(\xi)$ are called Hermite polynomials $H_n(\xi)$. The corresponding differential eqn is $\frac{d^2H_n}{d\xi^2} - 2\xi\frac{dH_n}{d\xi} + 2n H_n = 0$. The first few Hermite polynomials are $H_0(\xi) = 1$, $H_1(\xi) = 2\xi$, $H_2(\xi) = 4\xi^2 - 2$, etc.

$$\boxed{\varphi_n(x) = N_n H_n\left(\sqrt{\frac{m\omega}{\hbar}}\,x\right) e^{-\frac{m\omega}{2\hbar}x^2}}$$

($\because \xi = \frac{x}{a}$ and $a = \sqrt{\frac{\hbar}{m\omega}}$)

$N_n \rightarrow$ normalization constant.

To summarize :

$$\varphi_n(x) = N_n H_n\left(\sqrt{\frac{m\omega}{\hbar}}\, x\right) e^{-\frac{m\omega}{2\hbar}x^2} \quad , \; n = 0, 1, 2, \ldots$$

$$\varphi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2} \longrightarrow$$ ground state eigenfunction is Gaussian

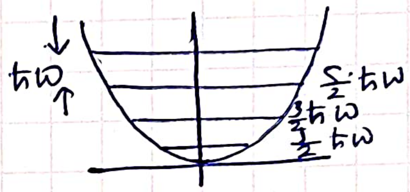
$$\varphi_1(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2}}\, 2 \sqrt{\frac{m\omega}{\hbar}} \cdot x \; e^{-\frac{m\omega}{2\hbar}x^2}$$

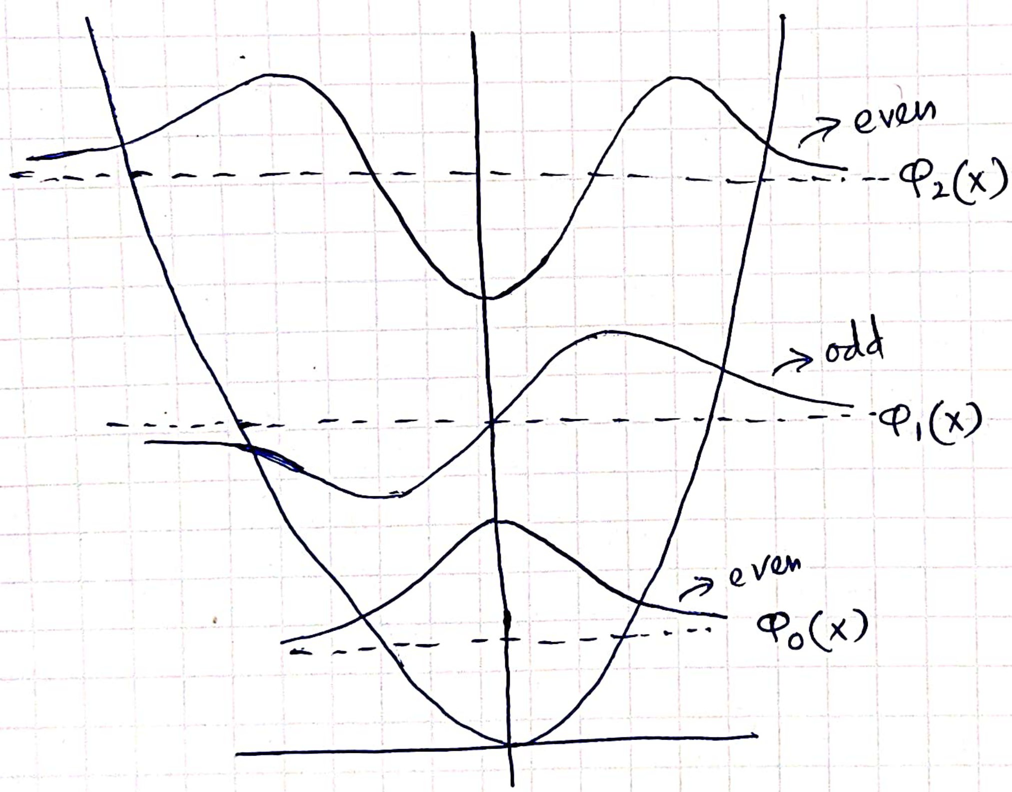
$$= 2\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \sqrt{\frac{m\omega}{2\hbar}}\; x\, e^{-\frac{m\omega}{2\hbar}x^2} \longrightarrow$$ 1st excited state odd function.

⋮

etc.

$$\boxed{E_n = \left(n + \tfrac{1}{2}\right)\hbar\omega}$$





Harmonic oscillator: Algebraic method

Write $\hat{H}$ as the product of operator $\hat{A}$ and $(\hat{A}^T)^* = \hat{A}^+$

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = \frac{1}{2}m\omega^2\left(\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2}\right)$$

Define $\hat{A} = \hat{x} + \frac{i\hat{p}}{m\omega} \Rightarrow \hat{A}^+\hat{A} = \left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)$

$$= \hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} + \frac{i}{m\omega}\underbrace{(\hat{x}\hat{p} - \hat{p}\hat{x})}_{i\hbar\mathbb{1}}$$

$$\Rightarrow \hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} = \hat{A}^+A + \frac{\hbar}{m\omega}\mathbb{1}$$

$$\Rightarrow \boxed{\hat{H} = \frac{1}{2}m\omega^2\hat{A}^+\hat{A} + \frac{1}{2}\hbar\omega\mathbb{1}}$$

Now, $[\hat{A}, \hat{A}^+] = \left[\hat{x} + \frac{i\hat{p}}{m\omega}, \hat{x} - \frac{i\hat{p}}{m\omega}\right] = \frac{-i}{m\omega}[\hat{x}, \hat{p}] + \frac{i}{m\omega}[\hat{p}, \hat{x}]$

$$= \frac{2\hbar}{m\omega}\mathbb{1}$$

$$\Rightarrow \left[\sqrt{\frac{m\omega}{2\hbar}}\hat{A}, \sqrt{\frac{m\omega}{2\hbar}}\hat{A}^+\right] = 1$$

We define

$$\boxed{\begin{aligned}\hat{a} &= \sqrt{\frac{m\omega}{2\hbar}}\hat{A} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)\\ \hat{a}^+ &= \sqrt{\frac{m\omega}{2\hbar}}\hat{A}^+ = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)\end{aligned}} \quad \boxed{[\hat{a}, \hat{a}^+] = 1}$$

Express $\hat{x}, \hat{p}$ in terms of $\hat{a}, \hat{a}^+$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^+)$$

$$\hat{p} = i\sqrt{\frac{m\omega\hbar}{2}}(\hat{a}^+ - \hat{a})$$

Now $\hat{a}^+a = \frac{m\omega}{2\hbar}\hat{A}^+\hat{A} \Rightarrow \hat{H} = \frac{1}{2}m\omega^2 \cdot \frac{2\hbar}{m\omega}\hat{a}^+\hat{a} + \frac{1}{2}\hbar\omega\mathbb{1}$

$$\Rightarrow \boxed{\hat{H} = \hbar\omega\left(\hat{a}^+\hat{a} + \frac{1}{2}\right)}$$

Define $\hat{N} = \hat{a}^+\hat{a}$ → called the number operator (Hermitian)

$$\boxed{\hat{H} = \left(\hat{N} + \frac{1}{2}\right)\hbar\omega}$$

$$\hat{H} = \left(\hat{N} + \frac{1}{2}\right)\hbar\omega \quad , \text{ where } \hat{N} = \hat{a}^{+}\hat{a}$$

What are the eigenstates of $\hat{N}$? $\hat{N}$ is Hermitian
$\Rightarrow$ real eigenvalues, $[\hat{N}, \hat{H}] = 0 \Rightarrow \hat{N}, \hat{H}$ have complete set of simultaneous eigenstates.

Let $\boxed{\hat{N}|n\rangle = n|n\rangle}$

n → eigenvalue, $|n\rangle$ → eigenstate of $\hat{N}$

Note: $[\hat{N}, \hat{a}] = [\hat{a}^{+}\hat{a}, \hat{a}] = \hat{a}^{+}[\hat{a}, \hat{a}] + [\hat{a}^{+}, \hat{a}]\hat{a} = -\hat{a}$

and $[\hat{N}, \hat{a}^{+}] = [\hat{a}^{+}\hat{a}, \hat{a}^{+}] = \hat{a}^{+}[\hat{a}, \hat{a}^{+}] + [\hat{a}^{+}, \hat{a}^{+}]\hat{a} = \hat{a}^{+}$

Now
$$\begin{aligned} N\hat{a}^{+}|n\rangle &= (\hat{N}\hat{a}^{+} - \hat{a}^{+}\hat{N} + \hat{a}^{+}\hat{N})|n\rangle \\ &= ([\hat{N}, \hat{a}^{+}] + \hat{a}^{+}\hat{N})|n\rangle \\ &= \hat{a}^{+}(\hat{N}+1)|n\rangle \qquad [\because [\hat{N}, \hat{a}^{+}] = \hat{a}^{+}] \\ &= \hat{a}^{+}(n+1)|n\rangle \\ &= (n+1)\hat{a}^{+}|n\rangle \end{aligned}$$

$$\Rightarrow \boxed{\hat{N}\hat{a}^{+}|n\rangle = (n+1)\hat{a}^{+}|n\rangle}$$

$\hat{a}^{+}|n\rangle$ is also an eigenstate of $\hat{N}$ with eigenvalue $(n+1)$

Similarly $\hat{N}\hat{a}|n\rangle = ([\hat{N}, \hat{a}] + \hat{a}\hat{N})|n\rangle = \hat{a}(\hat{N}-1)|n\rangle$

$$\Rightarrow \boxed{\hat{N}\hat{a}|n\rangle = (n-1)\hat{a}|n\rangle}$$

$\hat{a}|n\rangle$ is also an eigenstate of $\hat{N}$ with eigenvalue $(n-1)$

$\hat{a}^{+}$ applied on $|n\rangle$ creates 1 quantum unit of energy $\hbar\omega$
$\hat{a}^{+} \rightarrow$ creation / raising operator
$\hat{a}$ applied on $|n\rangle$ annihilates 1 quantum unit of energy $\hbar\omega$
$\hat{a} \rightarrow$ annihilation / lowering operator

Recall, $\langle\psi|\hat{a}^{+}a|\psi\rangle = ||a\psi||^2 \geq 0$, for any state $|\psi\rangle$

There exists a state $|0\rangle$ such that $a|0\rangle = 0$

$|0\rangle \rightarrow$ ground state

We can write $\hat{a}|n\rangle = c_n|n-1\rangle$

$$\langle n|\hat{a}^{+} = \langle n-1|c_n^*$$

$$\langle n|\hat{a}^{+}\hat{a}|n\rangle = \underbrace{\langle n-1|n-1\rangle}_{1 \text{ (normalized)}} c_n^* c_n = |c_n|^2$$

$\Rightarrow \quad |c_n|^2 = \langle n|\hat{N}|n\rangle = n \qquad [\because \hat{N}|n\rangle = n|n\rangle]$

$\Rightarrow \quad c_n = \sqrt{n}\, e^{2\varphi}$

We choose $\varphi = 0 \quad \Rightarrow \boxed{\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle}$

Similarly, $\boxed{\hat{a}^{+}|n\rangle = \sqrt{n+1}\,|n+1\rangle}$

Now $\hat{a}|n\rangle = \sqrt{n}\,|n-1\rangle$

$\Rightarrow \hat{a}^2|n\rangle = \hat{a}\sqrt{n}\,|n-1\rangle = \sqrt{n(n-1)}\,|n-2\rangle$

$\Rightarrow \hat{a}^3|n\rangle = \sqrt{n(n-1)(n-2)}\,|n-3\rangle$

$\vdots$

and so on ... till ... !!

Note that the sequence terminates if we start with a positive integer n.

n cannot be a non-integer $\Rightarrow$ the sequence must terminate with $n = 0$

n cannot be negative since $n = \langle n|\hat{N}|n\rangle = \langle n|\hat{a}^{+}\hat{a}|n\rangle \geq 0$

we define the ground state $|0\rangle$

$$\boxed{\hat{a}|0\rangle = 0}$$

Again, the first excited state $|1\rangle$ is given by,

$$|1\rangle = \hat{a}^{+}|0\rangle$$

subsequent excited states are obtained as,

$$|2\rangle = \frac{\hat{a}^{+}}{\sqrt{2}}|1\rangle = \frac{(\hat{a}^{+})^{2}}{\sqrt{2}}|0\rangle$$

$$|3\rangle = \frac{\hat{a}^{+}}{\sqrt{3}}|2\rangle = \left[\frac{(\hat{a}^{+})^{3}}{\sqrt{3!}}\right]|0\rangle$$

$$\vdots$$

$$\boxed{|n\rangle = \frac{(\hat{a}^{+})^{n}}{\sqrt{n!}}|0\rangle}$$

Also, $\hat{H}|n\rangle = \hbar\omega(\hat{a}^{+}a + \frac{1}{2})|n\rangle = \hbar\omega(\hat{N}+\frac{1}{2})|n\rangle$

$\Rightarrow$ $\boxed{\hat{H}|n\rangle = (n+\frac{1}{2})\hbar\omega|n\rangle}$

$\Rightarrow$ The energy eigenvalues : $\boxed{E_n = (n+\frac{1}{2})\hbar\omega}$

Eigenstates in position space

Ground state: If $\varphi_0(x)$ is the ground state in position space,

$$\hat{a}\,\varphi_0(x) = 0$$

$$\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)\varphi_0(x) = 0 \qquad \left[\because\ \hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} + \frac{i\hat{p}}{m\omega}\right)\right]$$

$$\Rightarrow\quad x\,\varphi_0(x) + \frac{i}{m\omega}\,\frac{\hbar}{i}\,\frac{\partial\varphi_0}{\partial x} = 0$$

$$\Rightarrow\quad \frac{d\varphi_0(x)}{dx} = -\frac{m\omega}{\hbar}\,x\,\varphi_0(x) \Rightarrow \varphi_0(x) = A_0\,e^{-\frac{m\omega}{2\hbar}x^2}$$

$$\int_{-\infty}^{\infty} dx\,|\varphi_0(x)|^2 = 1 \quad\Rightarrow\quad A_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$$

$$\boxed{\varphi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}}$$

1st excited state

$$|n\rangle = \frac{(\hat{a}^{+})^n}{\sqrt{n!}}|0\rangle , \quad \hat{a}^{+} = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)$$

$$|1\rangle = \hat{a}^{+}|0\rangle \Rightarrow \varphi_1(x) = \left(\hat{x} - \frac{i\hat{p}}{m\omega}\right)\varphi_0(x)\sqrt{\frac{m\omega}{2\hbar}}$$

$$= \left[x\,\varphi_0(x) - \frac{i}{m\omega}\frac{\hbar}{i}\frac{\partial}{\partial x}\varphi_0(x)\right]\sqrt{\frac{m\omega}{2\hbar}}$$

$$= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left[x e^{-\frac{m\omega}{2\hbar}x^2} - \frac{\hbar}{m\omega}\left(-\frac{m\omega}{2\hbar}\right) 2x\, e^{-\frac{m\omega}{2\hbar}x^2}\right]\sqrt{\frac{m\omega}{2\hbar}}$$

$$= 2\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\sqrt{\frac{m\omega}{2\hbar}}\; x\, e^{-\frac{m\omega}{2\hbar}x^2}$$

$$\Rightarrow \boxed{\varphi_1(x) = 2\left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\sqrt{\frac{m\omega}{2\hbar}}\; x\, e^{-\frac{m\omega}{2\hbar}x^2}}$$

In general $\boxed{\varphi_n(x) = \frac{1}{\sqrt{n!}}(\hat{a}^{+})^n\varphi_0(x)}$

$\varphi_n(x) \to$ a product of $e^{-\frac{m\omega}{2\hbar}x^2}$ and Hermite polynomial of degree n and "parity" $(-1)^n$

$$\varphi_n(x) = A_n\, e^{-\frac{m\omega}{2\hbar}x^2}\, H_n\left(\sqrt{\frac{m\omega}{2\hbar}}\,x\right)$$

Orthonormality criteria:

$$\int_{-\infty}^{\infty} dy\, e^{-y^2} H_n(y)\, H_m(y) = \underset{\text{const. factor}}{\underset{\downarrow}{c}}\,\delta_{mn}$$